

Overview of classical and quantum mechanics and its relevance to QUBIC

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Wiki references

- [Hamiltonian mechanics – Wikipedia](#)
 - [Lagrangian mechanics – Wikipedia](#)
 - [Action principles - Wikipedia](#)
 - [Path integral formulation – Wikipedia](#)
 - [Uncertainty principle - Wikipedia](#)
 - [Bohr model – Wikipedia](#)
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- Semiclassical theory: Reimers, Wilson, Heller J Chem Phys 79 (1983) P4749

Formulations of classical mechanics

- Newtonian formalism 1687

- $F = ma \quad a = \frac{d^2q}{dt^2} \quad F = -\frac{\partial V}{\partial q}$

F is the force, m is the mass, a the acceleration, q is the position, t is the time, V is the potential energy

- Hamiltonian formalism 1833 (used in MD codes)

$$H = T + V, \quad T = \frac{1}{2}mv^2 = \frac{p^2}{2m} \quad v = \frac{dq}{dt} \quad p = mv$$

H is the Hamiltonian, T is the kinetic energy, v is the velocity, p is the momentum

$$\frac{dp}{dt} = -\frac{\partial H}{\partial q} \quad \frac{dq}{dt} = \frac{\partial H}{\partial p}$$

- Lagrangian formalism 1760

$$L = T - V \quad \frac{d}{dt} \left(\frac{\partial L}{\partial \frac{dq}{dt}} \right) = \frac{\partial L}{\partial q}$$

L is the Lagrangian

Formalisms based on the classical action (functional)

- Definition of the action

$$S = \int_{t_1}^{t_2} L dt$$

t_1 and t_2 are the start and end times of a trajectory

S is the classical action

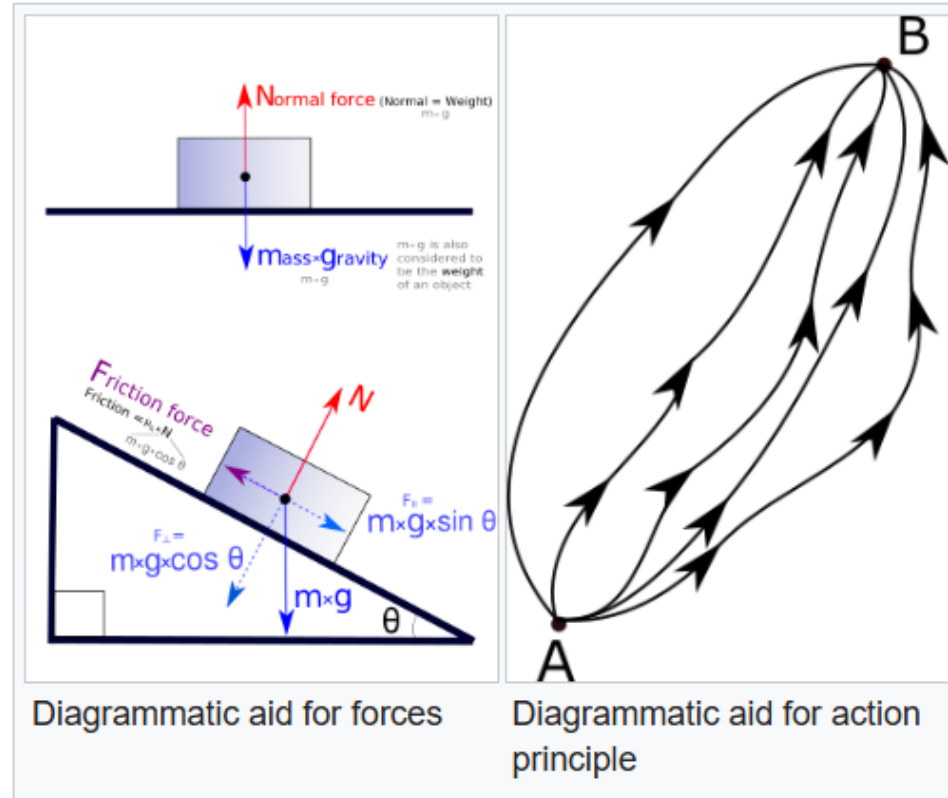
Thinking about the action means considering the whole path $C(q_t, p_t)$ as an entity rather than considering each point in time and asking what happens next.

You know the start point (q_{t_1}, p_{t_1}) and the end point (q_{t_2}, p_{t_2}) and ask about the properties of the interconnecting path

Paths, not points [\[edit \]](#)

The explanatory diagrams in force-based mechanics usually focus on a single point, like the [center of momentum](#), and show vectors of forces and velocities. The explanatory diagrams of action-based mechanics have two points with actual and possible paths connecting them.^[8] These diagrammatic conventions reiterate the different strong points of each method.

Action principles - Wikipedia



Depending on the action principle, the two points connected by paths in a diagram may represent two particle positions at different times, or the two points may represent values in a [configuration space](#) or in a [phase space](#). The mathematical technology and terminology of action principles can be learned by thinking in terms of physical space, then applied in the more powerful and general abstract spaces.

Action variational principle

- $(\delta S)_C = 0$

where $C(q_t, p_t)$ is the path

This means that if an infinitesimal change is made to a classical path, then there is no change in the action.

Emily Noether proved that any conserved quantity in the Lagrangian leads to symmetry property of the path, and visa versa.

Global symmetry in the path leads to the 7 conservation laws of classical mechanics.

Local symmetry leads to gauge theory and thence the modern theories of particle physics.

Protein sims for fixed endpoints, conserved energy:

- Maupertuis's principle: If the energy is conserved, then the classical path C taking the protein from one structure, q_1 , to another, q_2 , involves no variations of the **abbreviated action W**

$$(\delta W)_C = 0 \quad \text{where} \quad W = \int_{q_1}^{q_2} p dq$$

- Fermat's principle: for conserved systems at fixed energy E is related to the **action S** by

$$S = W - E\Delta t$$

where Δt is the time taken for the motion

Of all possible paths, the classical path is the one for which both S and Δt are minima.

Formulations of quantum mechanics

- Schrödinger equation, Heisenberg matrix mechanics, and Feynmann path integration
- In classical mechanics, the allowed path is the path with minimum action
- In quantum mechanics, all paths are possible, weighted by

$$e^{-i\Delta S/\hbar} = \cos \frac{\Delta S}{\hbar} - i \sin \frac{\Delta S}{\hbar}$$

where $\hbar = \frac{h}{2\pi}$ and by definition $h = 6.62607015 \times 10^{-34}$ Js

(note: speed of light is defined as 299,792,458 m/s; these definitions are used to generate the definitions of the meter and the kilogram in SI units)

$\hbar$ is called the quantum of action, also known as Planck's constant.

The contributing classical paths interfere with each other as the weightings can be both positive and negative.

Heisenberg uncertainty principle

- A quantum trajectory corresponds to many (interfering) paths
- Hence the action of a single classical trajectory can never describe the quantum result.
- Any measurement of the action therefore must embody some uncertainty:

error in any action measurement $\geq \hbar/2$

$$\Delta p \Delta q \geq \hbar/2$$

$$\Delta E \Delta t \geq \hbar/2$$

$$\Delta L \geq \hbar/2$$

where Δ indicated the standard deviation for measurements of energy E , time t , position q , momentum p , and angular momentum L

Some key results:

- Quantisation of the action in a radiation field (Planck equation):

$$S = n\hbar \quad \text{for } n = 0, 1, 2, 3 \dots \text{ leads to}$$
$$E = n\hbar\omega = nh\nu$$

- Approx. quantisation of the action in an electron in an atom (Bohr equation):

$$S = L = l\hbar \quad \text{for } l = 1, 2, 3 \dots n \quad (\text{actually } l = 0, 1, 2 \dots n - 1) \text{ leads to}$$

$$E_n = \frac{-1}{2n^2} \times \frac{k_e^2 e^4 m_e}{\hbar^2}$$

where k_e is Coulomb's constant, e and m_e are the electron charge and mass

- Quantisation of vibrational action (harmonic oscillator):

$$S = \left(n + \frac{1}{2}\right)\hbar \quad \text{for } n = 0, 1, 2, 3 \dots \text{ leads to}$$

$$E = \left(n + \frac{1}{2}\right)\hbar\omega = \left(n + \frac{1}{2}\right)h\nu$$

Correspondance principle as $\hbar \rightarrow 0$

- Quantum mechanics reduces to classical mechanics
- Easy: quantised energy levels for light and vibrators simply decay to zero

$$E = nh\nu \quad \text{or} \quad E = (n + \frac{1}{2})h\nu$$

- Tough: matter does not exist according to classical mechanics

$$E_n = \frac{-1}{2n^2} \times \frac{k_e^2 e^4 m_e}{\hbar^2}$$

- Nightmare: small changes in action become much more important, but statistical averaging cancels them all out !

$$e^{-i\Delta S/\hbar} = \cos \frac{\Delta S}{\hbar} - i \sin \frac{\Delta S}{\hbar}$$

20th century quantum science

- Tough problem: the Born-Oppenheimer approximation.
 - Electronic and nuclear motions separated
 - Provides the concept of a molecule as made up of interacting atoms
 - Electronic motion treated by quantum chemistry
 - Generate potential-energy surfaces for the understanding of nuclear motion
 - Both quantum and classical (MD, eg Alan Mark) treatments of nuclear motion
- Easy problem: Quantisation of optical and vibrational energy is a minor addition to classical mechanics
- Troubles with this: sometimes a full quantum description of electronic and nuclear motion is needed (e.g. Jeff, PsiQuantum) ... atomic structure concept lost
- Bigger picture: links to advanced quantum treatments of light, then to particle physics and cosmology

20th century quantum science

- Useful approximations account for:
 - Chemical structure and reactivity
 - Biochemistry, molecular biology, cell biology, etc.
 - Materials science, semiconductors, superconductors
 - Light-matter interactions (embodying much of photonics)
 - Light absorption, emission, lasing, Raman spectroscopy, Brillouin spectroscopy, NMR, upconversion, downconversion
- Much of QUBIC research is in this domain

21st century quantum science

- The nightmare problem of the path weightings:
$$e^{-i\Delta S/\hbar} = \cos \frac{\Delta S}{\hbar} - i \sin \frac{\Delta S}{\hbar}$$
- Quantum mechanics can be controlled by interference effects between different classical paths
- Thought to disprove quantum mechanics by Einstein etc, leads to Schrödinger's cat, basis for quantum computing.
- Generates entanglement, allowing more information to be stored in a quantum system than can be stored in a classical one.
- The information content becomes the same as $\hbar \rightarrow 0$, so how do we know what additional information we actually have?
- Quantum decoherence controls everything, need to measure and calculate it.
- **QUBIC related experiments with modelling challenges:**
 - Warwick Bowen, Alan Rowan cooling of specific protein/membrane vibrations
 - Sergey Kruk, Alina Kabakova: Paired photons may be entangled, and this could affect two-photon light-matter interactions
 - David Simpson: diamond NV centres with entangled spins
 - Lloyd Hollenberg using entanglement to make measurements

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Getting Quantum information from classical MD

- The trick is to use a formalism in which the quantum equations take the same form as the classical equations: Hamilton's equation
- We are interested in low frequency modes and so will use the classical temperature operator instead of the quantum one, so don't get scared.
- I give here the quantum equations and show how they reduce to the classical ones
- QUBIC is interested in non-equilibrium energy distributions, this is an easy addition
- I will give equations in terms of 1 degree of freedom only, but these readily generalise to include all possible motions of the atoms in a protein

Quantum thermal average in terms of eigenstates

- The quantum temperature operator is

$$\frac{e^{-\beta H}}{\text{Tr}(e^{-\beta H})}$$

where $\beta = \frac{1}{k_B T}$ with k_B as Boltzmann's constant, T the temperature, and Tr is the trace operator

- The value of any observable property operator A can be expressed as

$$\langle A \rangle = \text{Tr} \left(\frac{A e^{-\beta H}}{\text{Tr}(e^{-\beta H})} \right)$$

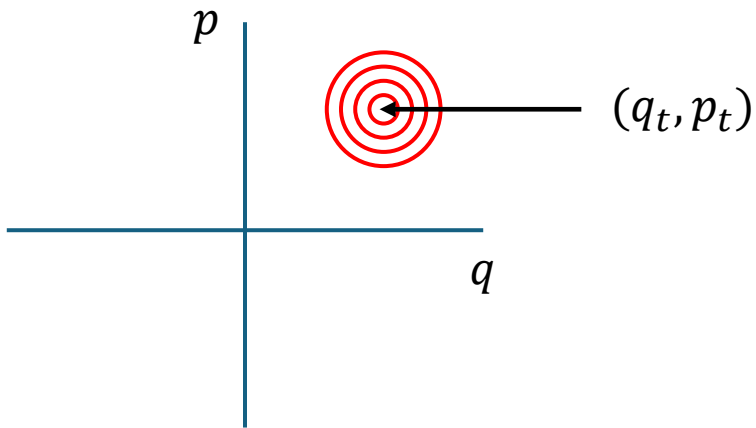
which can be expressed in terms of the eigenstates $|\psi_i\rangle$ of the system as

$$\langle A \rangle = \frac{\sum_i A_{ii} e^{-\beta E_i}}{\sum_i e^{-\beta E_i}}$$

Replace eigenstates with Gaussian wavepackets

- A gaussian wavepacket is a wavefunction that is:
 - is not an eigenstate, so it varies with time t
 - is expressed in terms of the position operator q and the momentum operator p ,
 - is centred in at position q_t and momentum p_t
 - has a phase γ_t
 - and a width α_t (spread in phase space so that it can obey the Heisenberg uncertainty principle)

$$|\psi_{q_t p_t}\rangle = \exp \left\{ \frac{i}{\hbar} [\alpha_t (q - q_t)^2 + p_t (q - q_t) + \gamma_t] \right\}$$



$$\langle q \rangle = q_t, \quad \langle p \rangle = p_t$$

Propagating a wavepacket forward in time

- The wavepacket advanced by time t under Hamiltonian operator H is

$$|\psi_{q_t p_t}\rangle = e^{-iHt/\hbar} |\psi_{q_0 p_0}\rangle$$

- If H is harmonic, then Gaussian wavepackets can change width

$$\frac{d\alpha_t}{dt} = -\frac{2\alpha_t^2}{m} - \frac{1}{2} \frac{\partial^2 V}{\partial q_t^2} \dots = 0 \text{ if } \alpha_t \text{ is set to match the force constant}$$

- and phase

$$\frac{d\gamma_t}{dt} = \frac{i\hbar\alpha_t}{m} - \frac{dS}{dt} \dots S \text{ is the classical action}$$

- with their centres evolving according to Hamilton's equation:

$$\frac{dp_t}{dt} = -\frac{\partial H}{\partial q_t} \quad \frac{dq_t}{dt} = \frac{\partial H}{\partial p_t}$$

- So there is a close relationship to classical mechanics
- For anharmonic potentials, assuming no change in wavepacket form:
 - Allowing the width to change with time is called the THAWED GAUSSIAN APPROXIMATION TGA
 - Constraining the width is called the FROZEN GAUSSIAN APPROXIMATION FGA

Quantum thermal average in terms of Gaussian wavepackets

$$\langle A \rangle = \frac{\int_{-\infty}^{\infty} dq_0 \int_{-\infty}^{\infty} dp_0 \langle \psi_{q_0 p_0} | A e^{-\beta H} | \psi_{q_0 p_0} \rangle}{\int_{-\infty}^{\infty} dq_0 \int_{-\infty}^{\infty} dp_0 \langle \psi_{q_0 p_0} | e^{-\beta H} | \psi_{q_0 p_0} \rangle}$$

- As $\hbar \rightarrow 0$, these Gaussian wavepackets collapse to delta functions at the classical points (q_0, p_0)

- This expression therefore collapses to

$$\langle A \rangle = \frac{\int_{-\infty}^{\infty} dq_0 \int_{-\infty}^{\infty} dp_0 A(q_0, p_0) e^{-\beta E(q_0, p_0)}}{\int_{-\infty}^{\infty} dq_0 \int_{-\infty}^{\infty} dp_0 e^{-\beta E(q_0, p_0)}}$$

where $E(q_0, p_0)$ is the total energy at that position and momentum.

- This is the **classical** equation used to determine properties using Monte Carlo simulations, which give the same results as does MD simulations

Preserving the form of the equations at finite $\hbar$

- Without approximation, the quantum temperature operator may be rewritten as a quantum time propagator:

$$\exp(-\beta H) = \exp\left(-\frac{iH\tau}{\hbar}\right) \text{ where } \tau \text{ is the imaginary time interval } \tau = \beta\hbar$$

- Hence the quantum temperature operator requires the solution of classical MD equations in imaginary time !
- We won't go there, just use the classical temperature approximation instead:

$$\langle A \rangle = \frac{\int_{-\infty}^{\infty} dq_0 \int_{-\infty}^{\infty} dp_0 e^{-\beta E(q_0, p_0)} \langle \psi_{q_0 p_0} | A | \psi_{q_0 p_0} \rangle}{\int_{-\infty}^{\infty} dq_0 \int_{-\infty}^{\infty} dp_0 e^{-\beta E(q_0, p_0)}}$$

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Calculating IR and Raman spectra

- For Raman, this applies to single-photon experiments only, not the paired photons used by Sergey Kruk, Irena Kabakova, etc.
- The quantum operator A involves $\frac{d\mu}{dq} q$ for IR, and $\frac{d\alpha}{dq} q$,
 - where μ is the dipole moment operator
 - and α is the polarizability operator.
- So to model Warwick's experiments, Callum will need to evaluate the polarizability derivatives, perhaps in Cartesian coordinates ?????

In both quantum and classical mechanics, 1-photon spectra can be expressed as a Fourier transform of an autocorrelation function

$$\sigma(\omega) = \int_{-\infty}^{\infty} C(t) e^{i\omega t} dt$$

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Evaluating the autocorrelation function

$$A = e^{-\beta H} e^{iHt/\hbar} \frac{d\mu}{dq} q e^{-iHt/\hbar} \frac{d\mu}{dq} q$$

- For constant $\frac{d\mu}{dq}$, the integrals $\langle \psi_{q_0 p_0} | A | \psi_{q_0 p_0} \rangle$ can be evaluated analytically as Gaussian integrals.
- Hence the autocorrelation function can be calculated by averaging over a classical thermal ensemble, and hence the spectrum.
- It seems to me that if the wavepacket widths don't change with time (Frozen Gaussian Approximation) and classical temperature is used, then the quantum spectrum reduced to the classical one Need to check.

Warwick's vibrational entanglement experiment

- Warwick wants to cool ONE protein or membrane long-range vibration (frequency order 50 cm^{-1}) down its zero-point level $|\psi^0\rangle$
- Then excite it to one quantum of vibration $|\psi^1\rangle$
- These states can then be entangled to make an initial state

$$|\psi_0\rangle = a|\psi_0^0\rangle + b|\psi_0^1\rangle$$

- The question is as to whether or not this quantum entangled state can remain entangled as the protein/membrane vibrates.

Modelling quantum coherence

- If it moves coherently, then the propagating wavefunction $|\psi_t\rangle$ will retain a simple relationship with the initial wavefunction $|\psi_0\rangle$, as told by the autocorrelation function

$$\begin{aligned} C(t) &= \langle \psi_0 | \psi_t \rangle = \langle \psi_0 | e^{-iHt/\hbar} | \psi_0 \rangle \\ &= a^2 \langle \psi_0^0 | \psi_t^0 \rangle + b^2 \langle \psi_0^1 | \psi_t^1 \rangle + ab \langle \psi_0^0 | \psi_t^1 \rangle + ab \langle \psi_0^1 | \psi_t^0 \rangle \end{aligned}$$

- There are 3 ways in which this function can decay with time:
 - T1: (classical) energy is transferred from the high-temperature surroundings to heat up the cooled mode ... **property of E_t for the cooled mode**
 - T2: (classical) the phase of the vibration averaged over different initial positions and momenta for the surroundings becomes randomised ... **property of (q_t, p_t)**
 - T2: (quantum) the wavepackets loose phase **property of γ_t (the classical action)**